22MAT230 MATHEMATICS FOR COMPUTING 4

***Batch : AIE -C***

***Team 7***

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**G Prajwal Priyadarshan - CB.SC.U4AIE24214**

**Kabilan K - CB.SC.U4AIE24224**

**Kishore B - CB.SC.U4AIE24227**

**Rahul L S - CB.SC.U4AIE24248**


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# LLM Guided Regularization of Pseudoinverse for Ill-Posed Signal Reconstruction

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## Problem Statement

## What Are ill Posed Inverse Problems?

A problem Ax = b is WELL POSED if:

1) Solution exists

2) Solution is unique

3) Solution depends continuously on data

We study conditioning using singular values and condition number.

## Well Posed vs ill Posed Problems

### Example

1) Full Rank Matrix

```
clc, clearvars
n = 100;
A1 = randn(n);              % Random Full rank matrix

[U1, S1, V1] = svd(A1);
singular_vals_A1 = diag(S1);

% Condition number
cond_A1 = cond(A1);
disp('Condtion number of Well Posed Matrix')
```

```
Condtion number of Well Posed Matrix
```

```
disp(cond_A1)
```

```
   5.3544e+03
```

2) ill Posed Matrix

```
function A = rankdeficient_operator(n, r)
    [U,~,V] = svd(randn(n), 'econ');   % random orthogonal matrices
    s = [linspace(1,0.1,r) zeros(1,n-r)];   % r non-zero singular values
    A = U * diag(s) * V';   % construct matrix
end
A2 = rank_deficient_operator(n, 12);
[U2, S2, V2] = svd(A2);
singular_vals_A2 = diag(S2);

% Condition number
cond_A2 = cond(A2);

disp('Condition number of Ill posed Matrix:')
```

```
Condition number of Ill posed Matrix:
```

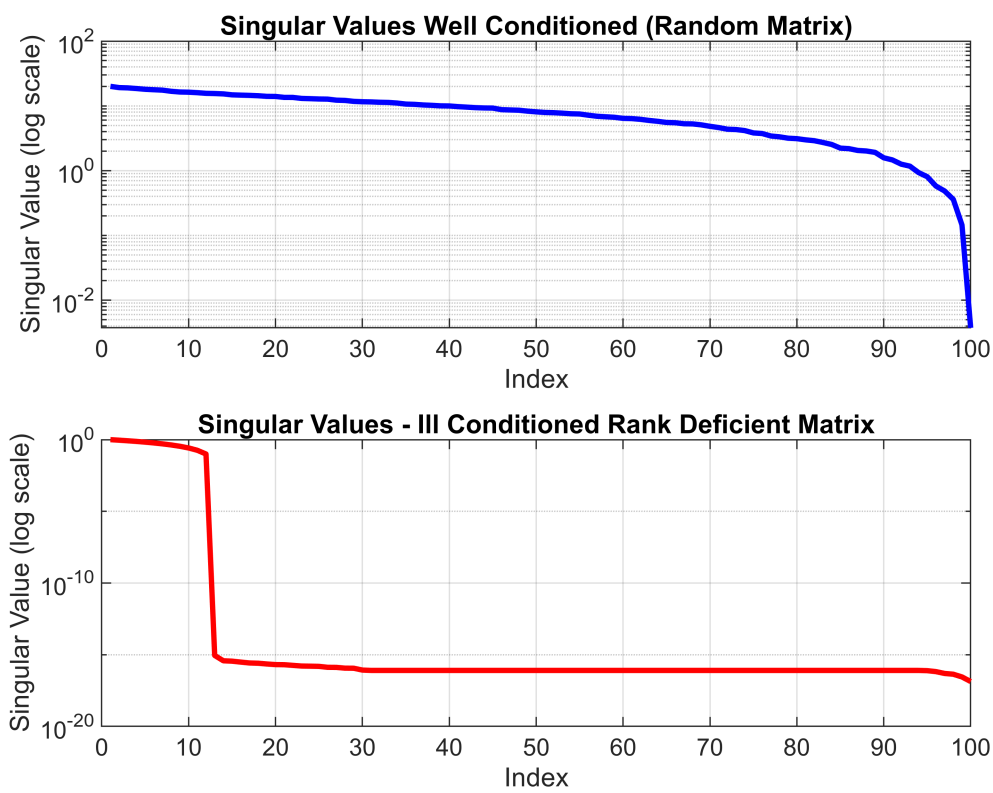
```
disp(cond_A2)
```

```
3.1797e+18
```

## Singular Value Spectrum

```
figure;
subplot(2,1,1)
semilogy(singular_vals_A1, 'b', 'LineWidth', 2);
grid on;
title('Singular Values Well Conditioned (Random Matrix)');
xlabel('Index');
ylabel('Singular Value (log scale)');

subplot(2,1,2)
semilogy(singular_vals_A2, 'r', 'LineWidth', 2);
grid on;
title('Singular Values - Ill Conditioned Rank Deficient Matrix');
xlabel('Index');
ylabel('Singular Value (log scale)');
```




## Stability Check

```
b1 = randn(n,1);
b1_noisy = b1 + 1e-6*randn(n,1);

x1_original = A1 \ b1;
x1_noisy    = A1 \ b1_noisy;
```

```
difference_well = norm(x1_original - x1_noisy);

x2_original = A2 \ b1;
```

```
Warning: Matrix is close to singular or badly scaled. Results may be inaccurate. RCOND =  3.154765e-20.
```

```
x2_noisy    = A2 \ b1_noisy;
```

```
Warning: Matrix is close to singular or badly scaled. Results may be inaccurate. RCOND =  3.154765e-20.
```

```
difference_ill = norm(x2_original - x2_noisy);

disp('Difference for Well-Conditioned Matrix:')
```

```
Difference for Well-Conditioned Matrix:
```

```
disp(difference_well)
```

```
   5.6937e-05
```

```

disp('Difference for Ill-Conditioned Matrix:')
```

```
Difference for Ill-Conditioned Matrix:
```

```
disp(difference_ill)
```

```
   4.1918e+12
```

### Condition Number via Singular Value

```
cond_manual_A1 = max(singular_vals_A1) / min(singular_vals_A1);
cond_manual_A2 = max(singular_vals_A2) / min(singular_vals_A2);

disp('Manual Condition Number (Random):')
```

```
Manual Condition Number (Random):
```

```
disp(cond_manual_A1)
```

```
   5.3544e+03
```

```


disp('Manual Condition Number (Hilbert):')
```

```
Manual Condition Number (Hilbert):
```

```
disp(cond_manual_A2)
```

```
   7.4082e+16
```

## Methodology

This study investigates inverse problem behavior under different numerical conditions. First, **synthetic signals** are generated to create controlled experimental data. A **forward model** is applied to simulate the measurement process, and **noise models** are introduced to represent real-world uncertainties. The **condition number** of the system matrix is analyzed to identify well-conditioned and ill-conditioned cases. Different **reconstruction methods** are then used to recover the original signal from noisy observations, and their performance is evaluated using suitable **evaluation metrics**.

## Signal Generation

```
function y = sinusoid(t)
    y = sin(2*pi*t) + 0.5*sin(6*pi*t);
end
```

```
function y = multisine(t, freqs, amps)
    if nargin < 2
        freqs = [2 5 9];
    end
    if nargin < 3
        amps = [1.0 0.6 0.3];
    end

    y = zeros(size(t));
    for k = 1:length(freqs)
        y = y + amps(k) * sin(2*pi*freqs(k)*t);
    end
end
```

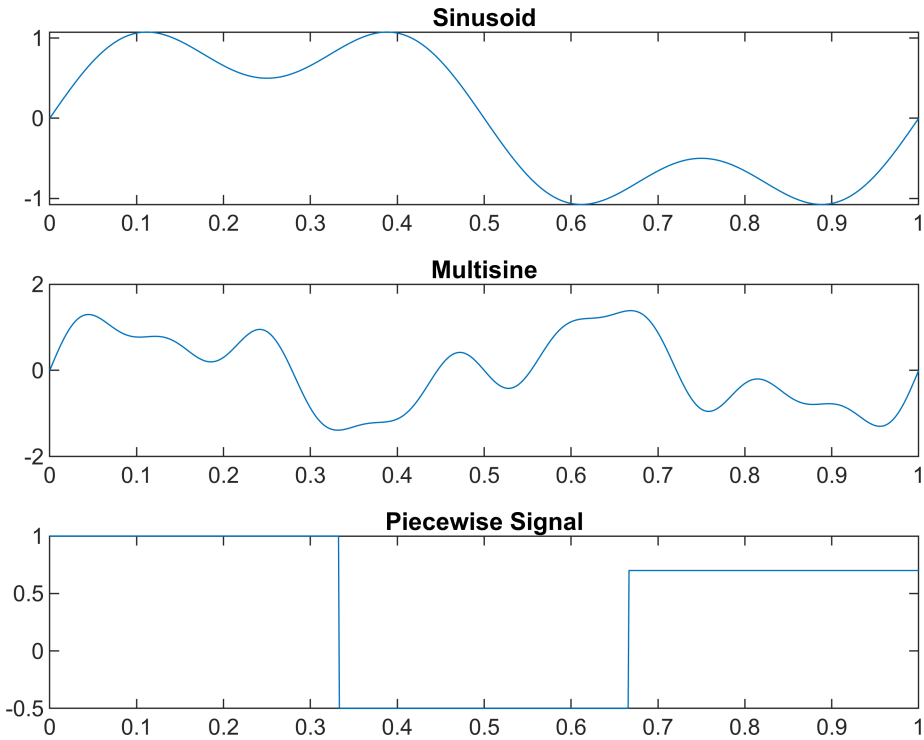
```
function y = piecewise_signal(t)
    y = zeros(size(t));
    N = length(t);
    thirds = floor(N/3);

    y(1:thirds) = 1.0;
    y(thirds+1:2*thirds) = -0.5;
    y(2*thirds+1:end) = 0.7;
end
```

```
t = linspace(0, 1, 1000);

y1 = sinusoid(t);
y2 = multisine(t);
y3 = piecewise_signal(t);

figure;
subplot(3,1,1); plot(t, y1); title('Sinusoid');
subplot(3,1,2); plot(t, y2); title('Multisine');
subplot(3,1,3); plot(t, y3); title('Piecewise Signal');
```




## Forward Models

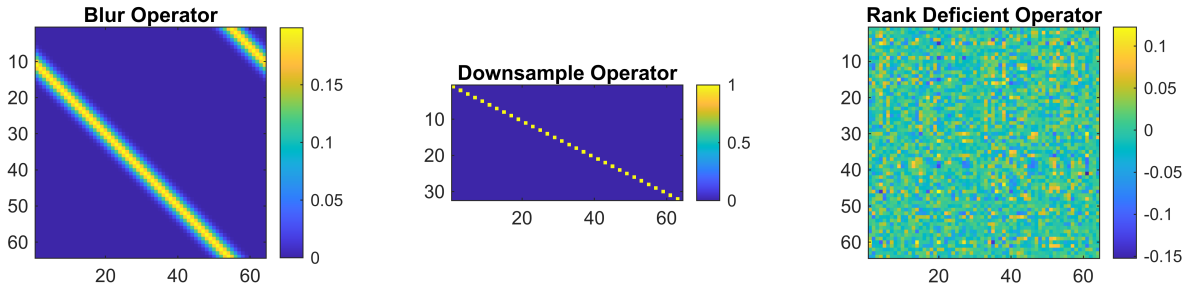
```
n = 100;

A_blur = blur_operator(n, 2.0);
A_down = downsample_operator(n, 2);
A_rank = rankdeficient_operator(n, 12);

figure('Position',[100 100 1000 300])
subplot(1,3,1); imagesc(A_blur); axis image;
title('Blur Operator'); colorbar;

subplot(1,3,2); imagesc(A_down); axis image;
title('Downsample Operator'); colorbar;

subplot(1,3,3); imagesc(A_rank); axis image;
title('Rank Deficient Operator'); colorbar;
```




```
function A = blur_operator(n, sigma, kernel_radius)
% BLUR_OPERATOR  Circulant Gaussian blur matrix

    if nargin < 3
        kernel_radius = 10;
    end

    ksize = 2 * kernel_radius + 1;
    k = gaussian_kernel(ksize, sigma);

    A = zeros(n, n);

    for i = 1:n
        for j = 1:n
            shift = mod(i - j, n);
            if shift < ksize
                A(i, j) = k(shift + 1);
            end
        end
    end
end
```

```
function A = downsample_operator(n, factor)
% DOWNSAMPLE_OPERATOR  Downsampling matrix

    if nargin < 2
        factor = 2;
    end

    m = floor(n / factor);
    A = zeros(m, n);

    for i = 1:m
        A(i, (i-1)*factor + 1) = 1.0;
    end
```

```
end
```

```
function A = rank_deficient_operator(n, r)
    [U,~,V] = svd(randn(n), 'econ');   % random orthogonal matrices
    s = [linspace(1,0.1,r) zeros(1,n-r)];   % r non-zero singular values
    A = U * diag(s) * V';   % construct matrix
end
```

```
function k = gaussian_kernel(sz, sigma)
% GAUSSIAN_KERNEL  1D Gaussian kernel

    idx = -(floor(sz/2)) : floor(sz/2);
    k = exp(-0.5 * (idx / sigma).^2);
    k = k / sum(k);
end
```

## Noise Models

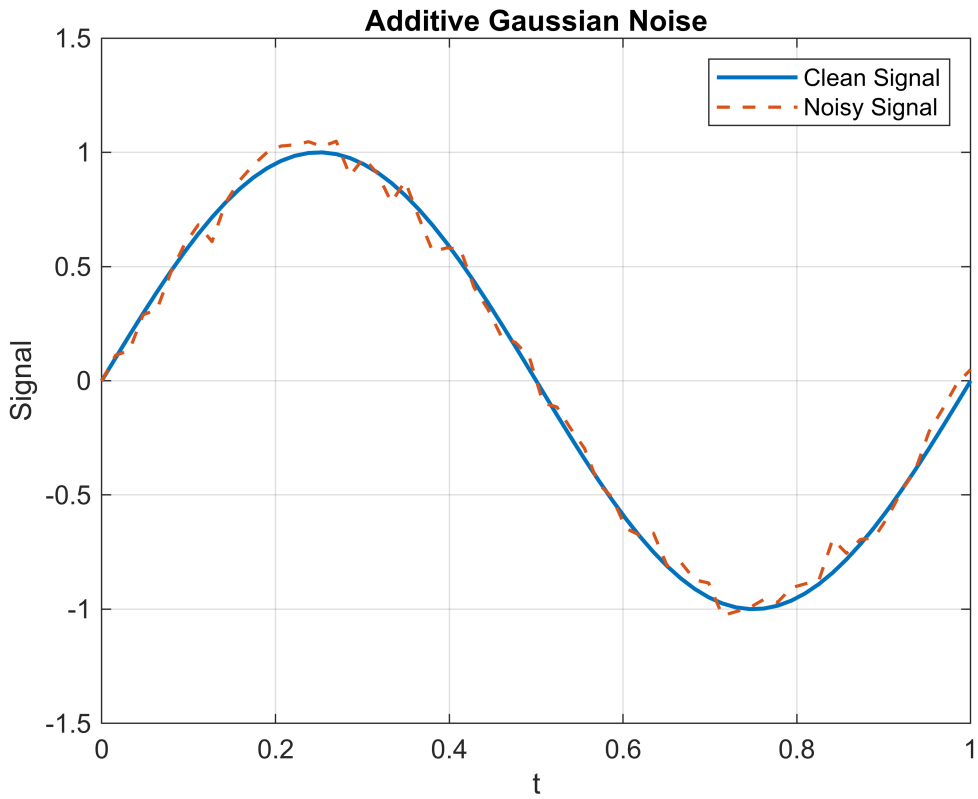
```
n = 100;
t = linspace(0,1,n)';
x = sin(2*pi*t);

sigma = 0.05;   % noise level

[y_noisy, noise] = add_gaussian_noise(x, sigma);

figure
plot(t,x,'LineWidth',1.5)
hold on
plot(t,y_noisy,'--','LineWidth',1.2)

legend('Clean Signal','Noisy Signal')
title('Additive Gaussian Noise')
xlabel('t')
ylabel('Signal')
grid on
```




```
function [y_noisy, noise] = add_gaussian_noise(y, sigma, rng)

    if nargin < 3 || isempty(rng)
        noise = sigma * randn(size(y));
    else
        noise = sigma * randn(rng, size(y));
    end

    y_noisy = y + noise;

end
```

## Reconstruction Methods

```
n = 100;                % signal length
noise_level = 0.08;     % noise standard deviation
blur_sigma = 2.5;       % blur kernel width

% Generate true signal
t = linspace(0,1,n);
x_true = sin(2*pi*t)' + 0.5*sin(6*pi*t)';

% Create forward operator (blur matrix)
A = blur_operator(n, blur_sigma, 10);
```

```
% Compute condition number
cond_num = cond(A);

fprintf('Forward operator matrix A: shape = %d x %d\n', size(A,1), size(A,2));
```

```
Forward operator matrix A: shape = 100 x 100
```

```
fprintf('Condition number (using diagnostics): κ(A) = %.2e\n', cond_num);
```

```
Condition number (using diagnostics): κ(A) = 1.94e+06
```

```


% Forward model
y_clean = A * x_true;

% Add Gaussian noise
[y_noisy, noise_vec] = add_gaussian_noise(y_clean, noise_level);

%From this we will use A and y_noisy for reconstruction
```

## PINV

```
function x = pseudoinverse_reconstruct(A, y_noisy)
    % Reconstruct signal using pseudoinverse
    % Inputs:
    % A - Forward operator matrix (n x m)
    % y_noisy - Noisy measurement vector (n x 1)

    % Output:
    % x - Reconstructed signal (m x 1)

    x = pinv(A) * y_noisy;
end

% Reconstruct
x_pinv = pseudoinverse_reconstruct(A, y_noisy);

% Compute reconstruction errors
mse_val = mse_error(x_true, x_pinv);
rel_error = relative_error(x_true, x_pinv);

fprintf('Relative Error: %.4f\n', rel_error);
```

```
Relative Error: 27107.9725
```

```
fprintf('MSE: %.6f\n', mse_val);
```

```
MSE: 454683593.547804
```

## TSVD

```
function x = tsvd_reconstruct(A, y, k)
    % TSVD_RECONSTRUCT  Truncated SVD reconstruction
    % Inputs:
    % A - Forward operator matrix
    % y - noisy observation
    % k - truncation parameter
    % Output:
    % x - reconstructed signal

    [U,S,V] = svd(A,'econ');

    % Truncate SVD components
    Uk = U(:,1:k);
    Sk = S(1:k,1:k);
    Vk = V(:,1:k);

    % TSVD solution
    x = Vk * (Sk \ (Uk' * y));
end
```

```
% Parameter sweep for TSVD
step = max(1, floor(n/50));
ks = 1:step:n-1;

num_k = length(ks);

mse_vals = zeros(num_k,1);
rel_vals = zeros(num_k,1);

x_store = cell(num_k,1);

for i = 1:num_k

    k = ks(i);

    % TSVD reconstruction
    x_tsvd = tsvd_reconstruct(A, y_noisy, k);

    % Metrics
    mse_vals(i) = mse_error(x_true, x_tsvd);
    rel_vals(i) = relative_error(x_true, x_tsvd);

    x_store{i} = x_tsvd;

end

% Find optimal k
[best_mse_tsvd, best_idx] = min(mse_vals);
best_k = ks(best_idx);
```

```
best_rel_tsvd = rel_vals(best_idx);
x_best_tsvd = x_store{best_idx};

fprintf('\nTSVD Parameter Sweep Results:\n');
```

```
TSVD Parameter Sweep Results:
```

```
fprintf('k range: %d to %d\n', ks(1), ks(end));
```

```
k range: 1 to 99
```

```
fprintf('Optimal k: %d\n', best_k);
```

```
Optimal k: 7
```

```
fprintf('Optimal MSE: %.6f\n', best_mse_tsvd);
```

```
Optimal MSE: 0.000948
```

```
fprintf('Optimal Relative Error: %.6f\n', best_rel_tsvd);
```

```
Optimal Relative Error: 0.039142
```

## Tikhonov

```
function x_tik = tikhonov_reconstruct(A, y, lambda)
% Tikhonov reconstruction
% Solves: min ||Ax - y||^2 + lambda ||x||^2

% Inputs:
% A - Forward operator matrix
% y - Observed noisy data
% lambda - Regularization parameter

% Output:
% x_tik - Reconstructed signal

[m, n] = size(A);

% Identity matrix
I = eye(n);

% Tikhonov solution
x_tik = (A' * A + lambda * I) \ (A' * y);
end
```

```
lambdas = logspace(-6,2,50);

mse_vals = zeros(length(lambdas),1);
rel_err_vals = zeros(length(lambdas),1);

x_results = cell(length(lambdas),1);
```

```
for i = 1:length(lambdas)

    lam = lambdas(i);

    % Reconstruction
    x_tik = tikhonov_reconstruct(A, y_noisy, lam);

    % Metrics
    mse_vals(i) = mse_error(x_true, x_tik);
    rel_err_vals(i) = relative_error(x_true, x_tik);

    x_results{i} = x_tik;

end

% Find optimal lambda
[best_mse_tikh, best_idx] = min(mse_vals);

best_lambda = lambdas(best_idx);
best_rel_tikh = rel_err_vals(best_idx);
x_best_tikh = x_results{best_idx};

fprintf('Tikhonov Parameter Sweep Results:\n');
```

```
Tikhonov Parameter Sweep Results:
```

```
fprintf('Lambda range: %.2e to %.2e\n', lambdas(1), lambdas(end));
```

```
Lambda range: 1.00e-06 to 1.00e+02
```

```
fprintf('Optimal lambda: %.6e\n', best_lambda);
```

```
Optimal lambda: 7.906043e-02
```

```
fprintf('Optimal MSE: %.6f\n', best_mse_tikh);
```

```
Optimal MSE: 0.008548
```

```
fprintf('Optimal Relative Error: %.6f\n', best_rel_tikh);
```

```
Optimal Relative Error: 0.117539
```

## NIST

```
function [x_rec, residuals, rel_errors, stop_iter] = nsit_morozov(A, y, x_true,
delta, q, tau, max_iter)

% NSIT reconstruction using Morozov discrepancy principle
% A        : forward operator
% y        : noisy measurements
% x_true   : true signal (only for error evaluation)
% delta    : noise level
```

```
% q        : geometric decay factor (0<q<1)
% tau      : Morozov safety factor
% max_iter : maximum iterations

[m,n] = size(A);

x = zeros(n,1);

At = A';
AtA = At*A;

% Proper scaling of alpha
alpha0 = norm(A)^2;

residuals = zeros(max_iter,1);
rel_errors = zeros(max_iter,1);

for k = 1:max_iter

    % Geometric alpha schedule
    alpha = alpha0 * q^(k-1);

    % Compute residual
    r = y - A*x;

    % NSIT correction step
    correction = (AtA + alpha*eye(n)) \ (At*r);

    % Update solution
    x = x + correction;

    % Residual norm
    residuals(k) = norm(y - A*x);

    % Relative error (for evaluation)
    rel_errors(k) = relative_error(x_true, x);

    % Morozov stopping rule
    if residuals(k) <= tau * delta
        break
    end

end

x_rec = x;
stop_iter = k;

residuals = residuals(1:k);
rel_errors = rel_errors(1:k);
```

```
end
```

```
max_iter = 200;
q = 0.9;
tau = 1.01;

delta = norm(noise_vec);

[x_nsit, residuals, errors, stop_iter] = nsit_morozov(A, y_noisy, x_true, delta, q,
tau, max_iter);

nsit_best_mse = mse_error(x_true,x_nsit);
nsit_best_rel = errors(end);

fprintf('NSIT stopped at iteration %d\n', stop_iter);
```

```
NSIT stopped at iteration 5
```

```
fprintf('Mean Squared Error = %.6f\n', nsit_best_mse);
```

```
Mean Squared Error = 0.003386
```

```
fprintf('Relative Error = %.6f\n', nsit_best_rel);
```

```
Relative Error = 0.073978
```

## FNIST

```
function x_rec = fnist_reconstruction(A, y, alpha0, q, max_iter)
    [~,n] = size(A);
    x = zeros(n,1);
    At = A';
    AtA = At*A;

    for k = 1:max_iter
        alpha = alpha0 * q^(k-1);
        r = y - A*x;
        % First correction
        F1 = (AtA + alpha*eye(n)) \ (At*r);
        % Second correction (correct form)
        temp = (AtA + alpha^2*eye(n)) \ (At*r);
        F2 = alpha * ((AtA + alpha*eye(n)) \ temp);
        % Stable update
        x = x + 0.7*F1 + 0.3*F2;
    end
    x_rec = x;
end

alpha0 = 1;
```

```
q = 0.9;
max_iter = 10;

x_fnist = fnist_reconstruction(A, y_noisy, alpha0, q, max_iter);

% metrics
mse_val = mse_error(x_true, x_fnist);
rel_error = relative_error(x_true, x_fnist);

fprintf('\nFNIST Reconstruction Results:\n');
```

```
FNIST Reconstruction Results:
```

```
fprintf('Relative Error: %.6f\n', rel_error);
```

```
Relative Error: 0.124670
```

```
fprintf('MSE: %.6f\n', mse_val);
```

```
MSE: 0.009617
```

```
alpha_vals = logspace(-2,2,25);   % alpha0 values
q_vals = 0.85:0.01:0.99;           % decay factors
iter_vals = 5:30;                 % iterations

best_mse = inf;
best_rel = inf;
best_alpha = 0;
best_q = 0;
best_iter = 0;
best_x = zeros(size(x_true));

for a = 1:length(alpha_vals)
    for b = 1:length(q_vals)
        for c = 1:length(iter_vals)

            alpha0 = alpha_vals(a);
            q = q_vals(b);
            max_iter = iter_vals(c);

            x_fnist = fnist_reconstruction(A, y_noisy, alpha0, q, max_iter);

            mse_val = mse_error(x_true, x_fnist);
            rel_err = relative_error(x_true, x_fnist);

            if mse_val < best_mse
                best_mse = mse_val;
                best_rel = rel_err;
                best_alpha = alpha0;
                best_q = q;
                best_iter = max_iter;
```

```
                best_x = x_fnist;
            end
        end
    end
end

best_mse_fnist = best_mse;
best_rel_fnist = best_rel;
x_fnist_best = best_x;

fprintf("\nOptimal FNIST Parameters:\n");
fprintf("alpha0 = %.4f\n", best_alpha);
fprintf("q = %.3f\n", best_q);
fprintf("iterations = %d\n", best_iter);
fprintf("Best MSE = %.6f\n", best_mse_fnist);
fprintf("Best Relative Error = %.6f\n", best_rel_fnist);
```

### Evaluation Metrics

```
function err = relative_error(x_true, x_rec)
err = norm(x_true - x_rec) / norm(x_true);
end

function mse = mse_error(x_true, x_rec)
mse = mean((x_true - x_rec).^2);
end
```

## Model Comparison

```
methods = {'Pseudoinverse','TSVD','Tikhonov','NSIT','FNIST'};

rel_errors = [
    relative_error(x_true, x_pinv);
    best_rel_tsvd;
    best_rel_tikh;
    nsit_best_rel;
    best_rel_fnist
];

mses = [
    mse_error(x_true, x_pinv);
    best_mse_tsvd;
    best_mse_tikh;
    nsit_best_mse;
    best_mse_fnist
];

fprintf('\n=====================================\n');
```

```
=====================================
```

```
fprintf('Reconstruction Method Comparison\n');
```

```
Reconstruction Method Comparison
```

```
fprintf('=====================================\n');
```

```
=====================================
```

```
for i = 1:length(methods)
    fprintf('%-12s | Rel Error: %.6f | MSE: %.6f\n',...
        methods{i}, rel_errors(i), mses(i));
end
```

```
Pseudoinverse | Rel Error: 27107.972469 | MSE: 454683593.547804
TSVD         | Rel Error: 0.039142 | MSE: 0.000948
Tikhonov     | Rel Error: 0.117539 | MSE: 0.008548
NSIT         | Rel Error: 0.073978 | MSE: 0.003386
FNIST        | Rel Error: 0.065126 | MSE: 0.002624
```

## LLM Parameter selection

```
% BUILD PROMPT FOR LLM

prompt = "";

prompt = prompt + "You are an expert in numerical linear algebra and inverse
problems." + newline;
prompt = prompt + "Your task is to recommend optimal regularization parameters for
solving an ill-posed inverse problem." + newline;
prompt = prompt + "We are solving the reconstruction problem Ax = y where A is
ill-conditioned and the measurements contain noise." + newline + newline;

prompt = prompt + "=====================================" + newline;
prompt = prompt + "PROBLEM DESCRIPTION" + newline;
prompt = prompt + "=====================================" + newline;

prompt = prompt + "Signal model:" + newline;
prompt = prompt + "- Signal type: multisine signal (sum of multiple sinusoids)" +
newline;
prompt = prompt + "- Signal length: " + num2str(n) + newline;
prompt = prompt + "- Sampling interval: uniform in [0,1]" + newline + newline;

prompt = prompt + "Forward model:" + newline;
prompt = prompt + "- Operator type: Gaussian blur operator" + newline;
prompt = prompt + "- Blur sigma: " + num2str(blur_sigma) + newline;
prompt = prompt + "- Matrix size: " + num2str(size(A,1)) + " x " +
num2str(size(A,2)) + newline;
prompt = prompt + "- Matrix rank: " + num2str(rank(A)) + newline;
prompt = prompt + "- Condition number: " + num2str(cond_num) + newline + newline;

prompt = prompt + "Noise model:" + newline;
```

```
prompt = prompt + "- Additive Gaussian noise" + newline;
prompt = prompt + "- Noise standard deviation: " + num2str(noise_level) + newline;
prompt = prompt + "- Noise norm: " + num2str(norm(noise_vec)) + newline + newline;

prompt = prompt + "======================================" + newline;
prompt = prompt + "CURRENT RECONSTRUCTION PERFORMANCE" + newline;
prompt = prompt + "======================================" + newline;

prompt = prompt + "Relative reconstruction errors:" + newline;
prompt = prompt + "- Pseudoinverse: " + num2str(rel_errors(1)) + newline;
prompt = prompt + "- TSVD: " + num2str(rel_errors(2)) + newline;
prompt = prompt + "- Tikhonov: " + num2str(rel_errors(3)) + newline;
prompt = prompt + "- NSIT: " + num2str(rel_errors(4)) + newline;
prompt = prompt + "- FNIST: " + num2str(rel_errors(5)) + newline + newline;

prompt = prompt + "Mean squared errors:" + newline;
prompt = prompt + "- Pseudoinverse: " + num2str(mses(1)) + newline;
prompt = prompt + "- TSVD: " + num2str(mses(2)) + newline;
prompt = prompt + "- Tikhonov: " + num2str(mses(3)) + newline;
prompt = prompt + "- NSIT: " + num2str(mses(4)) + newline;
prompt = prompt + "- FNIST: " + num2str(mses(5)) + newline + newline;

prompt = prompt + "======================================" + newline;
prompt = prompt + "OBJECTIVE" + newline;
prompt = prompt + "======================================" + newline;

prompt = prompt + "Select improved parameters that minimize reconstruction error
and improve stability." + newline;
prompt = prompt + "Consider the ill-conditioning of A and the noise level when
choosing parameters." + newline;
prompt = prompt + "Avoid parameter choices that cause instability or noise
amplification." + newline + newline;

prompt = prompt + "======================================" + newline;
prompt = prompt + "ALGORITHMS TO TUNE" + newline;
prompt = prompt + "======================================" + newline;

prompt = prompt + "1) TSVD: choose truncation parameter k" + newline;
prompt = prompt + "2) Tikhonov: choose regularization parameter lambda" + newline;
prompt = prompt + "3) NSIT: choose q, iterations, delta, tau" + newline;
prompt = prompt + "4) FNIST: choose alpha0, q, iterations" + newline + newline;

prompt = prompt + "======================================" + newline;
prompt = prompt + "PARAMETER CONSTRAINTS" + newline;
prompt = prompt + "======================================" + newline;

prompt = prompt + "- k must be between 5 and 30" + newline;
prompt = prompt + "- lambda between 1e-6 and 1" + newline;
prompt = prompt + "- alpha0 between 0.001 and 0.5" + newline;
prompt = prompt + "- q between 0.5 and 0.95" + newline;
```

```
prompt = prompt + "- iterations between 10 and 60" + newline;
prompt = prompt + "- tau between 1.0 and 1.2" + newline;
prompt = prompt + "- delta should be close to the noise norm" + newline + newline;

prompt = prompt + "=====================================" + newline;
prompt = prompt + "RESPONSE FORMAT" + newline;
prompt = prompt + "=====================================" + newline;

prompt = prompt + "Return ONLY valid JSON with no explanation." + newline;
prompt = prompt + "Use exactly this structure:" + newline;

prompt = prompt + '{' + newline;
prompt = prompt + ' "TSVD": {' + newline;
prompt = prompt + '   "k": 12' + newline;
prompt = prompt + ' },' + newline;
prompt = prompt + ' "Tikhonov": {' + newline;
prompt = prompt + '   "lambda": 0.015' + newline;
prompt = prompt + ' },' + newline;
prompt = prompt + ' "NSIT": {' + newline;
prompt = prompt + '   "q": 0.92,' + newline;
prompt = prompt + '   "iterations": 15,' + newline;
prompt = prompt + '   "delta": 0.1,' + newline;
prompt = prompt + '   "tau": 1.02' + newline;
prompt = prompt + ' },' + newline;
prompt = prompt + ' "FNIST": {' + newline;
prompt = prompt + '   "alpha0": 0.15,' + newline;
prompt = prompt + '   "q": 0.9,' + newline;
prompt = prompt + '   "iterations": 18' + newline;
prompt = prompt + ' }' + newline;
prompt = prompt + '}';
```

```
api_key = "AIzaSyCDPkmuUnzZpdr63Smj4g0eKHd8jrOMOO8";
params = get_llm_parameters_gemini(prompt, api_key);
k_llm = params.TSVD.k;

lambda_llm = params.Tikhonov.lambda;

q_nsit = params.NSIT.q;
iter_nsit = params.NSIT.iterations;
delta_nsit = params.NSIT.delta;
tau_nsit = params.NSIT.tau;

alpha_fnist = params.FNIST.alpha0;
q_fnist = params.FNIST.q;
iter_fnist = params.FNIST.iterations;

disp("LLM Suggested Parameters:")
```

```
LLM Suggested Parameters:
```

```
disp(params)
```

```
        TSVD: [1×1 struct]
    Tikhonov: [1×1 struct]
        NSIT: [1×1 struct]
       FNIST: [1×1 struct]
```

```
% RECONSTRUCTION WITH LLM PARAMETERS

% Tikhonov
x_rec_tikh_llm = tikhonov_reconstruct(A, y_noisy, lambda_llm);

% TSVD
x_rec_tsvd_llm = tsvd_reconstruct(A, y_noisy, k_llm);

% NSIT
[x_rec_nsit_llm,~,~,~] =
nsit_morozov(A,y_noisy,x_true,delta_nsit,q_nsit,tau_nsit,iter_nsit);

% FNIST
x_rec_fnist_llm = fnist_reconstruction( ...
    A, ...
    y_noisy, ...
    alpha_fnist, ...
    q_fnist, ...
    iter_fnist );

% COMPUTE ERRORS
rel_tsvd_llm  = relative_error(x_true, x_rec_tsvd_llm);
rel_nsit_llm  = relative_error(x_true, x_rec_nsit_llm);
rel_fnist_llm = relative_error(x_true, x_rec_fnist_llm);
rel_tikh_llm = relative_error(x_true, x_rec_tikh_llm);

mse_tsvd_llm  = mse_error(x_true, x_rec_tsvd_llm);
mse_nsit_llm  = mse_error(x_true, x_rec_nsit_llm);
mse_fnist_llm = mse_error(x_true, x_rec_fnist_llm);
mse_tikh_llm = mse_error(x_true, x_rec_tikh_llm);
```

```
%% ==========================================================
%% VISUALIZATION
%% ==========================================================

figure

subplot(3,2,1)
plot(t',x_true,'LineWidth',2)
hold on
plot(t',y_noisy,'--')
title("Original vs Observed")
```

```
legend("True","Observed")
grid on

subplot(3,2,2)
plot(t',x_true,'LineWidth',2)
hold on
plot(t',x_rec_tsvd_llm)
title("TSVD")
legend("True","TSVD","TSVD + LLM")
```

```
Warning: Ignoring extra legend entries.
```

```
grid on

subplot(3,2,3)
plot(t',x_true,'LineWidth',2)
hold on
plot(t',x_rec_tikh_llm)
title("Tikhonov")
legend("True","Tikhonov","Tikhonov + LLM")
```

```
Warning: Ignoring extra legend entries.
```

```
grid on

subplot(3,2,4)
plot(t',x_true,'LineWidth',2)
hold on
plot(t',x_rec_nsit_llm)
title("NSIT")
legend("True","NSIT","NSIT + LLM")
```

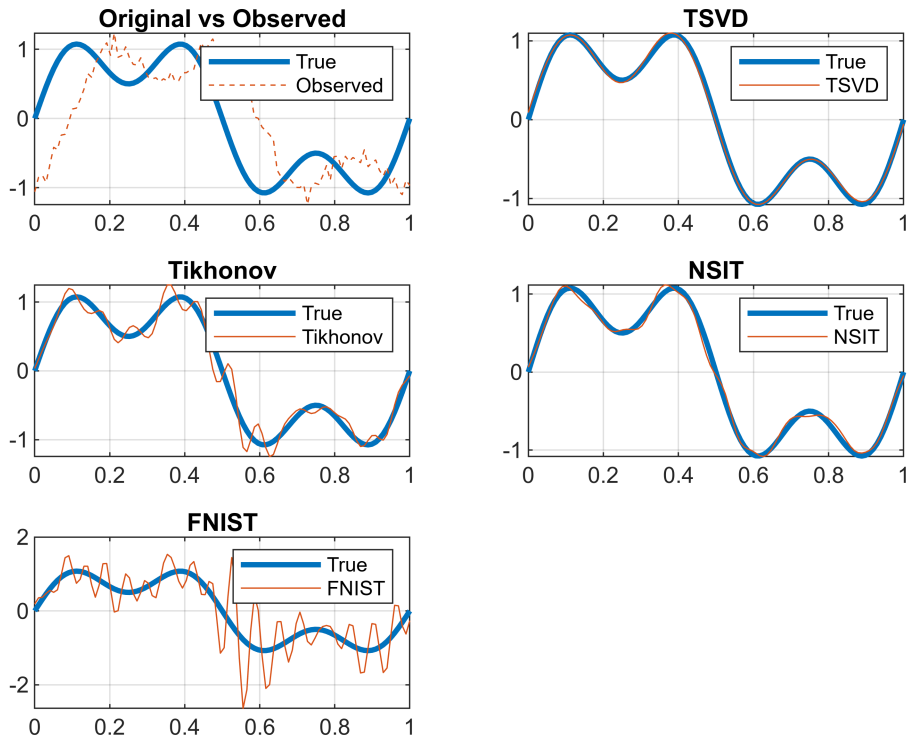
```
Warning: Ignoring extra legend entries.
```

```
grid on

subplot(3,2,5)
plot(t',x_true,'LineWidth',2)
hold on
plot(t',x_rec_fnist_llm)
title("FNIST")
legend("True","FNIST","FNIST + LLM")
```

```
Warning: Ignoring extra legend entries.
```

```
grid on
```




```
% PRINT FINAL RESULTS

fprintf("\n----- Reconstruction Errors -----\n")
```

```
----- Reconstruction Errors -----
```

```
fprintf("TSVD                : %.6f\n",best_rel_tsvd)
```

```
TSVD              : 0.039142
```

```
fprintf("TSVD + LLM          : %.6f\n",rel_tsvd_llm)
```

```
TSVD + LLM        : 0.041002
```

```
fprintf("Tikhonov            : %.6f\n",best_rel_tikh)
```

```
Tikhonov          : 0.117539
```

```
fprintf("Tikhonov + LLM      : %.6f\n",rel_tikh_llm)
```

```
Tikhonov + LLM    : 0.181156
```

```
fprintf("NSIT                : %.6f\n",nsit_best_rel)
```

```
NSIT              : 0.073978
```

```
fprintf("NSIT + LLM        : %.6f\n",rel_nsit_llm)
```

```
NSIT + LLM       : 0.073509
```

```
fprintf("FNIST             : %.6f\n",best_rel_fnist)
```

```
FNIST            : 0.065126
```

```
fprintf("FNIST + LLM       : %.6f\n",rel_fnist_llm)
```

```
FNIST + LLM      : 0.700339
```

```
function params = get_llm_parameters_gemini(prompt, api_key)

url = "https://generativelanguage.googleapis.com/v1/models/gemini-2.5-
flash:generateContent?key=" + api_key;
data = struct;
data.contents = {struct("parts",{struct("text",prompt)})};

options = weboptions( ...
    'MediaType','application/json', ...
    'ContentType','json', ...
    'Timeout',60);

response = webwrite(url,data,options);

text = response.candidates(1).content.parts(1).text;
% Remove markdown if present
text = erase(text,"```json");
text = erase(text,"```");

% Extract JSON safely
start_idx = find(text=='{',1,'first');
end_idx   = find(text=='}',1,'last');

json_text = text(start_idx:end_idx);

params = jsondecode(json_text);

end
```